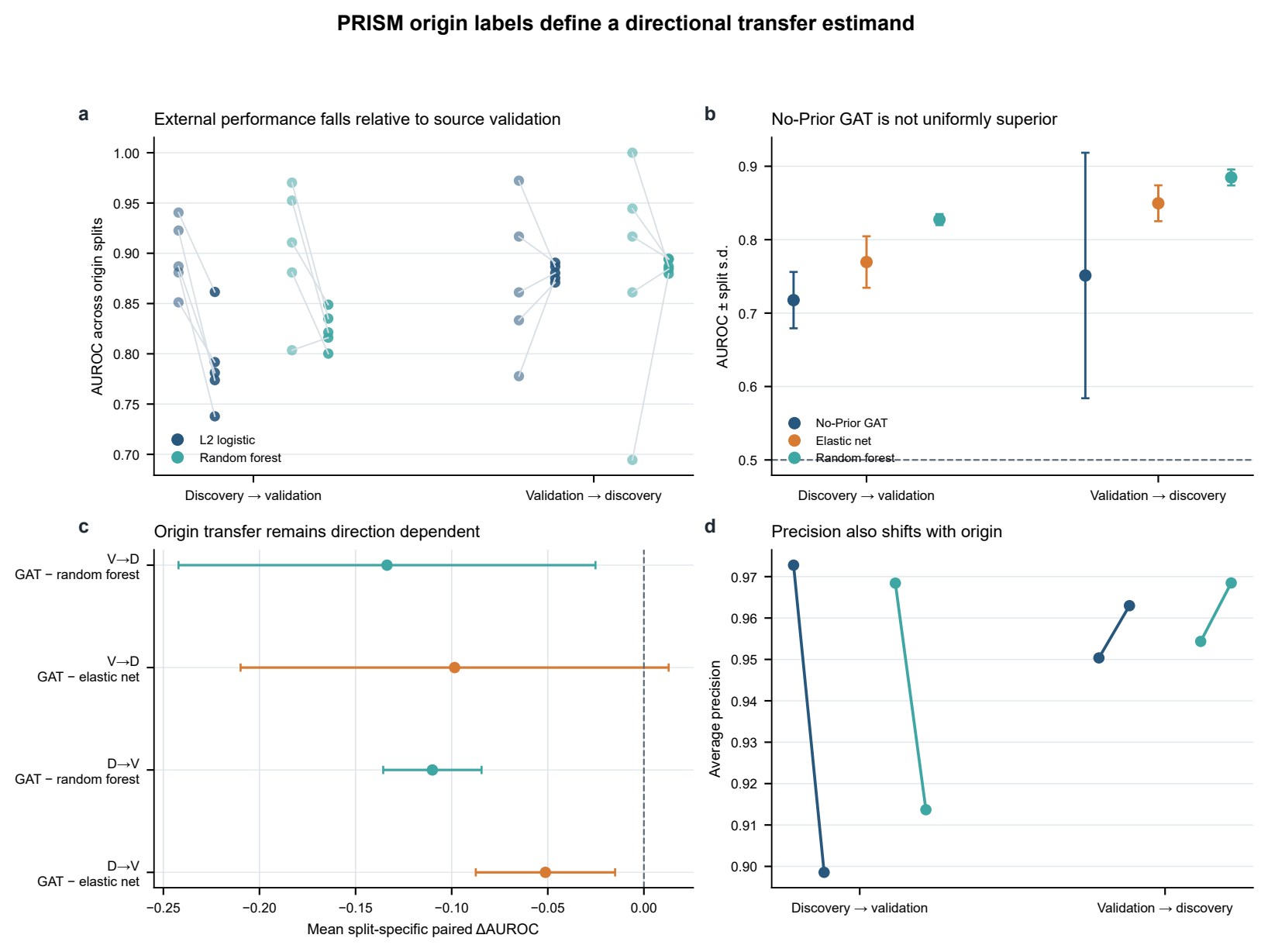

PRISM origin labels define a directional transfer estimand
a
External performance falls relative to source validation
AUROC across origin splits
1.00
0.95
0.90
0.85
0.80
0.75
0.70
L2 logistic
Random forest
Discovery → validation
Validation → discovery
b
No-Prior GAT is not uniformly superior
AUROC ± split s.d.
0.9
0.8
0.7
0.6
0.5
No-Prior GAT
Elastic net
Random forest
Discovery → validation
Validation → discovery
c
Origin transfer remains direction dependent
V→D
GAT − random forest
V→D
GAT − elastic net
D→V
GAT − random forest
D→V
GAT − elastic net
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Mean split-specific paired ΔAUROC
d
Precision also shifts with origin
Average precision
0.97
0.96
0.95
0.94
0.93
0.92
0.91
0.90
Discovery → validation
Validation → discovery